Shreya Nakum

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Education

University of California, Irvine Expected June 2027
Bachelor of Science in Computer Science with Honors (GPA: 3.80 / 4.00) *Irvine, CA*

- **Honors/Awards:** 1st Place Best Collaborative Research at UCI UROP 2025, Chancellor's Award for Excellence in Undergraduate Research, HackMIT Modal Honorable Mention, 6x Dean's Honors List.
- **Involvements:** Information and Computer Science Honors Program, Campus Honors Collegium, Artificial Intelligence@UCI

Technical Skills

Languages: Python, C++, C, R, Javascript/Typescript, Java, SQL, MIPS (Assembly), MATLAB, Fortran 77, Fortran 90.
Frameworks & Libraries: PyTorch, TensorFlow/Keras, Flask, FastAPI, BeautifulSoup, scikit-learn, transformers, Robot Operating System (ROS).
Tools & Platforms: AWS (Lambda, Boto3), Firebase, Docker, PostgreSQL, Milvus, HuggingFace, Git, WordPress, Simulink, Linux.
Expertise: Computer vision, CNNs, RAG, Data Analysis, Statistical Modeling, Physical Simulations, Academic Writing, Spatial Rotation.
Other: Web Development, API Integration, Web Scraping, MLOps, Supercomputing.

Experience

NASA: Johnson Space Center Jan 2026 – Aug 2026
AI/ML Engineering Intern *Houston, Texas*

- Implemented **reaction wheel actuator models** and developed an **attitude control system** with **MATLAB/Simulink**.
- Implemented MATLAB/Simulink models in a high-fidelity **attitude dynamics simulation** framework with **Fortran 77** for commercial partners' LEO space stations utilizing non-propulsive attitude control.
- Developed computer vision algorithm to detect green & red lasers in a dissimilar altimeter for the terminal descent of lander modules using ROS and OpenCV.

UC Irvine: Donald Bren School of Information & Computer Science Dec 2025 – Present
Researcher (Advisor: Dr. Krone-Martins) *Irvine, CA*

- Tested the OpenAI and Gemini embedding models against multiple datasets about mathematics and physics to identify where the model lacks in training data through graphical evaluation with PacMAP.
- Compared the models' manifolds using the Davies-Bouldin index, mean silhoette, and intra-cosine similiarity to determine their proclivity.

National Science Foundation: University of Oulu - M3S Lab June 2025 – Aug 2025
Researcher (Advisor: Dr. Matteo Esposito) *Oulu, North Ostrobothnia, Finland*

- Built a RAG pipeline to measure code clone similarity in software repositories.
- **Benchmarked 3 LLMs** across zero-shot, few-shot, and chain-of-through prompts with 3k+ thresholding values.
- Automated statistical evaluation of model metrics; co-authoring a research paper.

UC Irvine: Rose Labs (School of Biological Sciences) Sep 2024 – Present
Researcher (Advisor: Dr. Michael Rose) *Irvine, CA*

- Trained **220 supervised ML models** to identify evolving DNA sequences, achieving **80% accuracy**.
- Developed a **multi-camera tracking algorithm** implementing **2D pose estimation and identity propagation** for dense biological assays.
- Deployed a data annotation (**Flask+JS**) application that **accelerated data collection by 10x**.
- Designed a CNN object detection pipeline & trained a model (**95% accuracy**) to count *Drosophila melanogaster* eggs.
- Awarded 2 research prizes for novel small-object detection methods.

California Institute of Technology: Theoretical AstroPhysics Including Relativity and Cosmology Oct 2021 – Mar 2023
Researcher (Advisor: Dr. Iryna Butsky) *Pasadena, CA*

- Processed Hubble COS-Halos datasets with Python (**astropy, numpy, pandas**).
- Derived galaxy halo mass and velocity profiles; visualized results with **matplotlib.**
- Published findings in *MNRAS* and presented it at the American Astronomical Society conference.

Projects

HackMIT: ActsAI | *Honorable Mention, Modal.AI* GitHub

- Built a mental health platform with a Llama-based chatbot to connect users to therapists; integrated SMS alerts for crisis detection.

Atoma Media: Financial News Sentiment Analysis HuggingFace

- Built and automated Twitter scraper with PostgreSQL & AWS Lambda to collect financial news headlines and comments.
- Fine-tuned a sentiment-analysis model on financial news to predict market sentiment, achieving **80% accuracy and 6k+ downloads**.

Publications

Butsky, I., Nakum, S., Ponnada, S., Hummels, C., Ji, S., & Hopkins, P. (2022). Constraining Cosmic-ray Transport with Observations of the Circumgalactic Medium. *Monthly Notices of the Royal Astronomical Society*, **521**(2), 2477–2483. https://doi.org/10.1093/mnras/stad671

Nakum, S., Kemman, J., Esposito, M., Cerny, T., Lenarduzzi, V., & Taibi, D. (2025). Leveraging LLMs for Code Clone Similarity Classification in Open Source Microservice Repositories. *Automated Software Engineering*, **[In-Review]**

Nakum, S., & Qi, M. (2025). Drift and Anti-Drift: Hundreds of Generations of Evolutionary Stasis. Presented at the UC Irvine Undergraduate Research Symposium, Irvine, California

Nakum, S. (2026). Detecting LLM Knowledge Gaps. Presented at the UC Irvine Undergraduate Research Symposium, Irvine, California

Black, A., Nakum, S., Gooty, N., & Greenspan, Z. (2026). Drosophila Neuroassay. Presented at the UC Irvine Undergraduate Research Symposium, Irvine, California